

Exercise J01

Q1. The expression $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A}$ can be written as

- (A) $\sin A \cos A + 1$ (B) $\sec A \operatorname{cosec} A + 1$ (C) $\tan A + \cot A$ (D) $\sec A + \operatorname{cosec} A$

Ans. B

Sol.

$$\begin{aligned} & \frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A} \\ &= \frac{\sin A}{\cos A} \times \frac{1}{1 - \frac{\cos A}{\sin A}} + \frac{\cos A}{\sin A} \times \frac{1}{1 - \frac{\sin A}{\cos A}} \\ &= \frac{\sin A}{\cos A} \times \frac{\sin A}{\sin A - \cos A} + \frac{\cos A}{\sin A} \times \frac{\cos A}{\cos A - \sin A} \\ &= \frac{\sin^2 A}{\cos A (\sin A - \cos A)} - \frac{\cos^2 A}{\sin A (\sin A - \cos A)} \\ &= \frac{1}{(\sin A - \cos A)} \left[\frac{\sin^3 A - \cos^3 A}{\cos A \sin A} \right] \\ &= \frac{(\sin A - \cos A) (\sin^2 A + \sin A \cos A + \cos^2 A)}{(\sin A - \cos A) \cos A \sin A} \\ &= \frac{1 + \sin A \cos A}{\cos A \sin A} \quad (\because \sin^2 A + \cos^2 A = 1) \\ &= \sec A \operatorname{cosec} A + 1. \end{aligned}$$

Q2. Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$ where $x \in \mathbb{R}$ and $k \geq 1$. Then $f_4(x) - f_6(x)$ equals :

- (A) $1/4$ (B) $1/12$ (C) $1/6$ (D) $1/3$

Ans. B

Sol.

$$\begin{aligned} f_k(x) &= \frac{1}{k} \{\sin^k x + \cos^k x\} \\ \therefore f_4(x) - f_6(x) &= \frac{1}{4} \{\sin^4 x + \cos^4 x\} - \frac{1}{6} \{\sin^4 x + \cos^6 x\} \\ &= \frac{1}{4} \{1 - 2\sin^2 x \cos^2 x\} - \frac{1}{6} \{1 - 3\sin^2 x \cos^2 x\} = \frac{1}{12} \end{aligned}$$

Q3. If $a \cos \theta + b \sin \theta = 3$ and $a \sin \theta - b \cos \theta = 4$, then the value of $a^2 + b^2$ is :

- (A) 9 (B) 16 (C) 25 (D) None of these

Ans. C

Sol. -

Q4. The expression $\sec^2 \theta \cdot \operatorname{cosec}^2 \theta$ is equal to :

- (A) $\sin^2 \theta + \cos^2 \theta$ (B) $\sin^2 \theta - \cos^2 \theta$ (C) $\sec^2 \theta + \operatorname{cosec}^2 \theta$ (D) None of these

Ans. C

Sol.

$$\sec^2 \theta \operatorname{cosec}^2 \theta = \frac{1}{\sin^2 \theta \cos^2 \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta}$$

$$= \sec^2 \theta + \operatorname{cosec}^2 \theta$$

Q5. If $\tan 50^\circ - \tan 40^\circ = k \tan 10^\circ$, then the value of k is :

- (A) 2 (B) 1 (C) 3 (D) None of these

Ans. A

Sol.

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\Rightarrow \tan 50^\circ - \tan 40^\circ = \tan(50^\circ - 40^\circ) (1 + \tan 50^\circ \tan 40^\circ)$$

$$\therefore k = 1 + \tan 50^\circ \tan 40^\circ = \frac{\cos 40^\circ \cos 50^\circ + \sin 40^\circ \sin 50^\circ}{\cos 40^\circ \cos 50^\circ}$$

$$= \frac{2 \cos(50^\circ - 40^\circ)}{2 \cos 40^\circ \cos 50^\circ} = \frac{2 \cos(10^\circ)}{\cos(40^\circ + 50^\circ) + \cos(40^\circ - 50^\circ)}$$

$$\Rightarrow K = \frac{2 \cos 10^\circ}{0 + \cos 10^\circ} = 2$$

Q6. The exact value of $\frac{\sin 24^\circ \cos 6^\circ - \sin 6^\circ \sin 66^\circ}{\sin 21^\circ \cos 39^\circ - \cos 51^\circ \sin 69^\circ}$ is :

- (A) 0 (B) 1 (C) -1 (D) None of these

Ans. C

Sol. -

Q7. If $\tan \alpha + \cot \alpha = a$, then the value of $\tan^4 \alpha + \cot^4 \alpha$ is :

- (A) $a^2 - 4a + 2$ (B) $a^4 - 4a^2 + 2$ (C) $a^4 + 4a^2 - 2$ (D) None of these

Ans. B

Sol. Given $\tan \alpha + \cot \alpha = a$

$$\tan^4 \alpha + \cot^4 \alpha = (\tan^2 \alpha + \cot^2 \alpha)^2 - 2 \tan^2 \alpha \cot^2 \alpha = [(\tan \alpha + \cot \alpha)^2 - 2]^2 - 2$$

$$= (a^2 - 2)^2 - 2$$

$$= a^4 - 4a^2 + 4 - 2$$

$$= a^4 - 4a^2 + 2$$

Q8. The simplified value of, $\frac{2(\sin 2\theta + 2 \cos^2 \theta - 1)}{\cos \theta - \sin \theta - \cos 3\theta + \sin 3\theta}$ is :

(A) $\sin \theta$ (B) $\cos \theta$ (C) $\operatorname{cosec} \theta$ (D) $\sec \theta$

Ans. C

Sol. -

Q9. If $\cos \theta = \frac{1}{2} \left(a + \frac{1}{a} \right)$, then the value of $\cos 3\theta$ in terms of 'a' is :

(A) $\frac{1}{4} \left(a^3 + \frac{1}{a^3} \right)$ (B) $\frac{1}{2} \left(a^3 + \frac{1}{a^3} \right)$ (C) $4 \left(a^3 + \frac{1}{a^3} \right)$

(D) None of these

Ans. B

Sol.

$$\cos \theta = \frac{1}{2} \left(a + \frac{1}{a} \right)$$

$$\Rightarrow \cos 3\theta = 4\cos^3 \theta - 3\cos \theta$$

$$= \cos \theta (4\cos^2 \theta - 3)$$

$$= \left(a + \frac{1}{a} \right) \frac{1}{2} \left[4 \left(\frac{1}{2} \left(a + \frac{1}{a} \right) \right)^2 - 3 \right]$$

$$= \frac{1}{2} \left(a + \frac{1}{a} \right) \left[a^2 + \frac{1}{a^2} + 2 - 3 \right]$$

$$= \frac{1}{2} \left(a + \frac{1}{a} \right) \left(a^2 + \frac{1}{a^2} - 1 \right)$$

$$= \frac{1}{2} \left(a^3 + \frac{1}{a} - a + a + \frac{1}{a^3} - \frac{1}{a} \right) = \frac{1}{2} \left(a^3 + \frac{1}{a^3} \right)$$

Q10. $(1 + \tan \alpha \tan \beta)^2 + (\tan \alpha - \tan \beta)^2$ is equal to :

(A) $\tan^2 \alpha + \tan^2 \beta$ (B) $\cos^2 \alpha \cos^2 \beta$ (C) $\sec^2 \alpha \sec^2 \beta$ (D) $\tan^2 \alpha \tan^2 \beta$

Ans. C

Sol. $(1 + \tan \alpha \tan \beta)^2 + (\tan \alpha - \tan \beta)^2$

$$= 1 + \tan^2 \alpha \tan^2 \beta + 2 \tan \alpha \tan \beta + \tan^2 \alpha + \tan^2 \beta - 2 \tan \alpha \tan \beta$$

$$= 1 + \tan^2 \alpha + \tan^2 \beta (1 + \tan^2 \alpha)$$

$$= (1 + \tan^2 \alpha) (1 + \tan^2 \beta)$$

$$= \sec^2 \alpha \cdot \sec^2 \beta$$

Q11. If $\frac{1 + \sin 2x}{1 - \sin 2x} = \cot^2(a + x) \forall x \in \mathbb{R} - \left(n\pi + \frac{\pi}{4}\right), n \in \mathbb{N}$. Then 'a' equals :

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) None of these

Ans. C

Sol. Given

$$\frac{1 + \sin 2x}{1 - \sin 2x} = \cot^2(a + x) \forall x \in \mathbb{R} - \left\{n\pi + \frac{\pi}{4}\right\}$$

$$\Rightarrow \frac{\sin^2 x + \cos^2 x + 2 \sin x \cos x}{\sin^2 x + \cos^2 x - 2 \sin x \cos x} = \cot^2(a + x)$$

$$\Rightarrow \left[\frac{\sin x + \cos x}{\sin x - \cos x} \right]^2 = \cot^2(a + x)$$

$$\Rightarrow \left(\frac{1 + \tan x}{\tan x - 1} \right)^2 = \frac{1}{\tan^2(a + x)}$$

$$\Rightarrow \tan^2\left(\frac{\pi}{4} + x\right) \cdot \tan^2(a + x) = 1$$

$$\Rightarrow \left[\tan\left(\frac{\pi}{4} + x\right) \cdot \tan(a + x) \right]^2 = 1$$

$$\Rightarrow a = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$$

Q12. If $\tan \alpha = \frac{5}{6}$ and $\tan \beta = \frac{1}{11}$, then $\left(0 < \alpha, \beta < \frac{\pi}{2}\right)$:

- (A) $\alpha + \beta = \frac{\pi}{6}$ (B) $\alpha + \beta = \frac{\pi}{4}$ (C) $\alpha + \beta = \frac{\pi}{3}$ (D) None of these

Ans. B

Sol. Given $\tan \alpha = \frac{5}{6}$ $\tan \beta = \frac{1}{11}$ $\left(0 < \alpha, \beta < \frac{\pi}{2}\right)$

$$\Rightarrow \tan(\alpha + \beta) = \frac{\frac{5}{6} + \frac{1}{11}}{1 - \frac{5}{66}} = \frac{61}{61} = 1$$

$$\Rightarrow \alpha + \beta = \frac{\pi}{4} \quad \{0 < \alpha + \beta < \pi\}$$

Q13. If $\tan x \tan y = a$ and $x + y = \frac{\pi}{6}$, then $\tan x$ and $\tan y$ satisfy the equation :

- (A) $x^2 - \sqrt{3}(1 - a)x + a = 0$ (B) $\sqrt{3}x^2 - (1 - a)x + a\sqrt{3} = 0$

(C) $x^2 + \sqrt{3}(1-a)x - a = 0$ (D) $\sqrt{3}x^2 + (1+a)x - a\sqrt{3} = 0$

Ans. B

Sol. Given

$$x + y = \frac{\pi}{6} \text{ \& tan x tan y = a}$$

Let equation whose roots will be tan x & tan y is

$$t^2 - (\tan x + \tan y)t + \tan x \tan y = 0$$

$$t^2 - (\tan(x+y)(1 - \tan x \tan y))t + a = 0$$

$$\Rightarrow t^2 - \left(\tan\left(\frac{\pi}{6}\right) \cdot (1-a) \right)t + a = 0$$

$$\Rightarrow t^2 - \left(\frac{1-a}{\sqrt{3}} \right)t + a = 0$$

$$\Rightarrow \sqrt{3}t^2 - (1-a)t + a\sqrt{3} = 0$$

$$\Rightarrow \text{equation is } \sqrt{3}x^2 - (1-a)x + \sqrt{3}a = 0$$

Q14. Which of the following number(s) is/are rational ?

(A) $\sin 15^\circ$ (B) $\cos 15^\circ$ (C) $\sin 15^\circ \cos 15^\circ$ (D) $\sin 15^\circ \cos 75^\circ$

Ans. C

Sol. -

Q15. If $\cos x + \cos y + \cos \alpha = 0$ and $\sin x + \sin y + \sin \alpha = 0$, then $\cot\left(\frac{x+y}{2}\right) =$

(A) $\sin \alpha$ (B) $\cos \alpha$ (C) $\cot \alpha$ (D) $2 \sin \alpha$

Ans. C

Sol. -

- Q16. If the roots of the quadratic equation, $x^2 + px + q = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$ respectively, then the value of $2 + q - p$ is :
- (A) 2 (B) 3 (C) 0 (D) 1

Ans. B

Sol. $x^2 + px + q = 0$
 $\tan 30^\circ + \tan 15^\circ = -p$
 $\tan 30^\circ \cdot \tan 15^\circ = q$
 $\tan 45^\circ = \frac{\tan 30^\circ + \tan 15^\circ}{1 - \tan 30^\circ \tan 15^\circ} = \frac{-p}{1 - q} = 1$
 $\Rightarrow -p = 1 - q$
 $\Rightarrow q - p = 1 \quad \therefore 2 + q - p = 3$

- Q17. Let $\cos(\alpha + \beta) = \frac{4}{5}$ and let $\sin(\alpha - \beta) = \frac{5}{13}$, where $0 \leq \alpha, \beta \leq \frac{\pi}{4}$. Then $\tan 2\alpha =$
- (A) $\frac{25}{16}$ (B) $\frac{56}{33}$ (C) $\frac{19}{12}$ (D) $\frac{20}{7}$

Ans. B

Sol. Hint: $\tan 2\alpha = \tan [(\alpha + \beta) + (\alpha - \beta)]$

- Q18. In a ΔPQR , if $3 \sin P + 4 \cos Q = 6$ and $4 \sin Q + 3 \cos P = 1$, then the angle R is equal to :-
- (A) $\frac{3\pi}{4}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$

Ans. C

Sol. $3 \sin P + 4 \cos Q = 6$ (1)
 $4 \sin Q + 3 \cos P = 1$ (2)
 Squaring (1) and (2). Then adding $9 + 16 + 24 \sin(P + Q) = 37$
 $\Rightarrow 24 \sin(P + Q) = 12$
 $\Rightarrow \sin(P + Q) = \frac{1}{2} \Rightarrow \sin R = \frac{1}{2}$
 $\therefore R = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$
 But $R = \frac{5\pi}{6}$ doesn't satisfy the given equation so $R = \frac{\pi}{6}$

- Q19. If $3 \sin \beta = \sin(2\alpha + \beta)$ then :
- (A) $\tan(\alpha + \beta) = 2 \tan \alpha$ (B) $2 \sin \beta = \sin(\alpha - \beta) \cos \alpha$

(C) $\sin \beta = \cos (\alpha + \beta) \cos \alpha$ (D) none of these

Ans. A

Sol. -

Q20. If $\sin (x + 3\alpha) = 3 \sin (\alpha - x)$, then :

(A) $\tan x = \tan \alpha$ (B) $\tan x = \tan^2 \alpha$ (C) $\tan x = \tan^3 \alpha$ (D) $\tan x = 3 \tan \alpha$

Ans. C

Sol. $\sin (x + 3\alpha) = 3 \sin (\alpha - x)$

$$\Rightarrow \frac{\sin (x + 3\alpha)}{\sin (\alpha - x)} = 3$$

Apply componendo and dividendo

$$\Rightarrow \frac{\sin (x + 3\alpha) + \sin (\alpha - x)}{\sin (x + 3\alpha) - \sin (\alpha - x)} = \frac{4}{2} = 2$$

$$\Rightarrow \frac{2 \sin (2\alpha) \cos (x + \alpha)}{2 \cos (2\alpha) \sin (x + \alpha)} = 2$$

$$\Rightarrow 2 \tan (x + \alpha) = \tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\Rightarrow \left(\frac{\tan x + \tan \alpha}{1 - \tan x \tan \alpha} \right) \cdot 2 = 2 \left(\frac{\tan \alpha}{1 - \tan^2 \alpha} \right)$$

$$\tan \alpha - \tan x \tan^2 \alpha = \tan x + \tan \alpha - \tan^2 \alpha \tan x - \tan^3 \alpha$$

$$\Rightarrow \tan x = \tan^3 \alpha$$

Q21. If $\cos x = \frac{2 \cos y - 1}{2 - \cos y}$, where $x, y \in (0, \pi)$. Then $\tan \frac{x}{2} \cdot \cot \frac{y}{2}$ is equal to :

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{\sqrt{3}}$

Ans. B

Sol. $\frac{1}{\cos \alpha} = \frac{2 - \cos \beta}{2 \cos \beta - 1}$ Applying C/D $\Rightarrow \frac{1 - \cos \alpha}{1 + \cos \alpha} = \frac{3(1 - \cos \beta)}{1 + \cos \beta}$

$$\Rightarrow \tan^2 = 3 \tan^2 \frac{\beta}{2} \Rightarrow \tan^2 \frac{\alpha}{2} \cot^2 \frac{\beta}{2} = 3$$

Q22. $\left(1 + \cos \frac{\pi}{8}\right) \left(1 + \cos \frac{3\pi}{8}\right) \left(1 + \cos \frac{5\pi}{8}\right) \left(1 + \cos \frac{7\pi}{8}\right)$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{\pi}{8}$ (C) $\frac{1}{8}$ (D) $\frac{1 + \sqrt{2}}{2\sqrt{2}}$

Ans. C

Sol. -

Q23. $\cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots \cos 179^\circ =$

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol. -

Q24. If $A = \sin^2 x + \cos^4 x$, then for all real x :

- (A) $1 \leq A \leq 2$ (B) $\frac{3}{4} \leq A \leq \frac{13}{16}$ (C) $\frac{3}{4} \leq A \leq 1$ (D) $\frac{13}{16} \leq A \leq 1$

Ans. C

Sol. $A = \sin^2 x + \cos^4 x$

$$= \cos^4 x - \cos^2 x + 1$$

$$= (\cos^2 x)^2 - 2 \cdot \cos^2 x \cdot \frac{1}{2} + \frac{1}{4} - \frac{1}{4} + 1$$

$$= (\cos^2 x - \frac{1}{2})^2 + \frac{3}{4} \Rightarrow 0 \leq \left(\cos^2 x - \frac{1}{2} \right)^2 \leq \frac{1}{4}$$

$$0 \leq \cos^2 x \leq 1$$

$$\Rightarrow -\frac{1}{2} \leq \cos^2 x - \frac{1}{2} \leq \frac{1}{2}$$

$$\Rightarrow \frac{3}{4} \leq A \leq 1$$

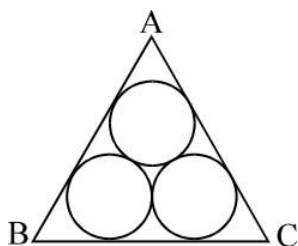
Q25. The sum of $\sin^3 \theta + \frac{1}{3} \sin^3 3\theta + \frac{1}{3^2} \sin^3 9\theta + \dots$ to n terms is :

- (A) $\frac{1}{2} \left[3 \sin \theta - \frac{1}{3^{n-1}} \sin(3^n \theta) \right]$ (B) $\frac{1}{2} \left[3 \cos \theta - \frac{1}{3^{n-1}} \sin(3^n \theta) \right]$
 (C) $\frac{1}{4} \left[3 \cos \theta - \frac{1}{3^{n-1}} \sin(3^n \theta) \right]$ (D) $\frac{1}{4} \left[3 \sin \theta - \frac{1}{3^{n-1}} \sin(3^n \theta) \right]$

Ans. D

Sol. -

Q26. In an equilateral triangle, 3 coins of radii 1 unit each are kept so that they touch each other and also the sides of the triangle. Height of the triangle is



- (A) $2 + 2\sqrt{3}$ (B) $3 + \sqrt{3}$ (C) $\sqrt{3}$ (D) $2 + \sqrt{3}$

Ans. B

Sol. The Height can be broken into Three Parts:

From A to Centre of Top Triangle + From Centre of Top Triangle to the Point of contact of bottom two circles + From Point of contact of bottom two circles to the base

$$= 2 + \sqrt{3} + 1 = 3 + \sqrt{3}$$

Q27. Let $\theta \in (0, \pi/4)$ and $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$, then

(A) $t_1 > t_2 > t_3 > t_4$ (B) $t_4 > t_3 > t_1 > t_2$ (C) $t_3 > t_1 > t_2 > t_4$ (D) $t_2 > t_3 > t_1 > t_4$

Ans. B

Sol. $\therefore \theta \in \left(0, \frac{\pi}{4}\right) \Rightarrow \tan \theta < 1 \text{ \& } \cot \theta > 1$

let $\tan \theta = 1 - x$ and $\cot \theta = 1 + y$

when $x, y > 0$ and are very small

$$\therefore t_1 = (1 - x)^{1-x}, t_2 = (1 - x)^{1+y}$$

$$t_3 = (1 + x)^{1-x}, t_4 = (1 + x)^{1+y}$$

clearly $t_4 > t_3 \text{ \& } t_1 > t_2$

also $t_3 > t_1$

Then $t_4 > t_1 > t_2$

Q28. A circle is inscribed inside a regular pentagon and another circle is circumscribed about this pentagon. Similarly a circle is inscribed in a regular heptagon and another circumscribed about the heptagon. The area of the regions between the two circles in two cases are A_1 and A_2 respectively. If each polygon has a side length of 2 units then which one of the following is true?

(A) $A_1 = \frac{5}{7}A_2$ (B) $A_1 = \frac{25}{49}A_2$ (C) $A_1 = \frac{49}{25}A_2$ (D) $A_1 = A_2$

Ans. D

Sol. -

Q29. If θ and ϕ are acute angles satisfying $\sin\theta = \frac{1}{2}$, $\cos\phi = \frac{1}{3}$, then $\theta + \phi \in$:

(A) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$ (B) $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ (C) $\left(\frac{2\pi}{3}, \frac{5\pi}{6}\right)$ (D) $\left(\frac{5\pi}{6}, \pi\right)$

Ans. B

Sol. Given

$$\sin \theta = \frac{1}{2}; 0 < \theta < \frac{\pi}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

$$\text{Given } \cos \phi = \frac{1}{3}; 0 < \phi < \frac{\pi}{2}$$

$$\Rightarrow \phi \in \left(\frac{\pi}{3}, \frac{\pi}{2} \right)$$

$$\Rightarrow \theta + \phi \in \left(\frac{\pi}{6} + \frac{\pi}{3}, \frac{\pi}{6} + \frac{\pi}{2} \right)$$

$$\Rightarrow \theta + \phi \in \left(\frac{\pi}{2}, \frac{2\pi}{3} \right)$$